Exploiting the Variance Risk Premium: A Systematic Credit Spread Strategy on SPY

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Abstract

This paper examines the variance risk premium (VRP), the persistent gap between options-implied volatility and subsequently realized volatility, and tests whether it can be profitably exploited through defined-risk credit spread strategies on the SPDR S&P 500 ETF (SPY). Using approximately 16 years of daily SPY price and VIX data (February 2010 through December 2025, 4,002 trading days), I document a mean VRP of 3.67%, with implied volatility exceeding realized volatility on 85.3% of all trading days. A systematic bull put credit spread strategy conditioned on a minimum VRP threshold of 3% achieves an 86.0% win rate across 121 monthly trades, with an annualized return of 37.5% and an annualized Sharpe ratio of 2.47. The VRP filter meaningfully reduces the maximum portfolio drawdown (24.31% versus 40.71% for the unfiltered strategy) at the cost of fewer trades. Tail risk analysis across five major stress events confirms that the VRP typically reverses during market dislocations, highlighting the strategy's primary vulnerability. A practical deployment framework is included for retail investors using defined-risk structures with limited capital.

1. Introduction

A well-documented pattern in equity options markets is that implied volatility consistently exceeds the volatility that actually materializes, a gap known as the variance risk premium (VRP). The intuition is straightforward: investors systematically overpay for options as insurance against tail risk, much like how most people pay more for home insurance than the expected value of their claims. This overpayment is rational from the buyer's perspective, but it creates a persistent structural edge for disciplined options sellers.

The existence and persistence of the VRP has significant implications for options sellers. If implied volatility is systematically inflated relative to realized volatility, then strategies that sell options and collect the inflated premium should generate positive expected returns over time. This insight underlies a broad family of 'short volatility' strategies employed by hedge funds, quantitative trading firms, and sophisticated retail traders.

However, short volatility strategies carry substantial tail risk. The same insurance-like properties that make options expensive in tranquil markets cause them to spike dramatically during crashes. The most well-known example is the February 2018 'Volmageddon' event, in which a sudden volatility spike wiped out several exchange-traded products that were mechanically short volatility. More recently, the COVID-19 pandemic drove the VIX to 82.7 on March 16, 2020, inflicting severe losses on undiscriminating short-vol positions.

This paper addresses a core question: can a simple, quantitatively-motivated filter based on the magnitude of the VRP itself improve the risk-adjusted performance of a short volatility credit spread strategy while retaining its fundamental edge? I argue that entering trades only when the premium is sufficiently elevated provides a natural filter against low-quality signals while preserving exposure to the best opportunities.

1.1 Motivation and Scope

The goal here is two-fold. On the academic side, I want to quantify the VRP using real market data and understand how it behaves across different regimes. On the practical side, I want to build something that could actually be traded with limited capital, using defined-risk structures that cap the maximum loss rather than requiring institutional-scale margin.

Unlike naked option selling, which requires substantial margin and carries theoretically unlimited downside, credit spreads (specifically bull put spreads) bound the maximum loss at the spread width minus the premium collected. This makes the strategy accessible with a few thousand dollars of capital, which is something a student or recent grad can realistically deploy.

2. Background & Related Theory

2.1 The Black-Scholes Framework and Implied Volatility

The Black-Scholes-Merton (BSM) model, foundational to modern options pricing, assumes constant volatility throughout the life of an option. Under this framework, an option's price is a deterministic function of the underlying asset price, strike price, time to expiration, risk-free rate, and volatility. In practice, traders invert the BSM formula to extract the implied volatility (IV) from observed market prices. This implied volatility represents the market's forward-looking expectation of price fluctuations.

If the BSM model were exactly correct, implied and realized volatility should converge in expectation. The systematic gap between them reveals that the market prices in a risk premium beyond what is statistically warranted, not because the model is wrong, but because volatility uncertainty itself has a price. Investors pay to offload the risk that volatility will be higher than expected, and that payment is what options sellers collect.

2.2 The Variance Risk Premium: Empirical Evidence

The VRP has been extensively documented in the academic literature. Bollerslev, Tauchen, and Zhou (2009) show that the VRP predicts future equity returns at monthly horizons, with a higher premium indicating either elevated risk aversion or heightened uncertainty about future volatility. The intuition is straightforward: when investors are particularly fearful, they overpay for downside protection, causing implied vol to become even more inflated relative to subsequent realized vol.

The VIX index, published by the CBOE, provides a standardized measure of 30-day implied volatility for the S&P 500 options market. Comparing the VIX at a given time to the realized volatility that unfolds over the subsequent 30 days yields the most commonly studied version of the VRP. Historically, the VRP has averaged roughly 3 to 6 percentage points, with the VIX exceeding subsequent realized vol roughly 75 to 85 percent of the time.

2.3 Credit Spreads: Defined-Risk Short Volatility

A bull put credit spread involves simultaneously selling an out-of-the-money (OTM) put and buying a further OTM put at a lower strike on the same underlying and expiration. The position receives a net credit and profits if the underlying remains above the short

put strike at expiration. The maximum profit is the credit received; the maximum loss is bounded by the spread width minus the credit.

This structure directly monetizes the VRP: the short put is overpriced due to elevated implied vol, and the long put hedge reduces both margin requirements and tail exposure. For a $500 underlying with a 2% wide spread, the capital at risk is approximately $650 to generate $350 in premium, a 53.8% return on risk if the trade succeeds. This capital efficiency is a key advantage over covered calls or cash-secured puts.

3. Data

The analysis uses daily closing price data for SPY and the VIX index, obtained via the yfinance library from Yahoo Finance. The sample spans from February 2010 through December 2025, comprising 4,002 trading days (~16 years). SPY provides the underlying equity price series; the VIX, published by the CBOE, serves as the daily measure of 30-day implied volatility for the S&P 500 options market.

Realized volatility is estimated as the rolling 21-trading-day standard deviation of SPY log returns, annualized by multiplying by the square root of 252. The VRP on each day is then:

$$VRPt = IVt - RVt$$

where IVt is the VIX closing value and RVt is the trailing 21-day realized volatility in annualized percentage terms.

The sample spans several major volatility regimes. The VIX reached a peak of 48.0 on August 8, 2011, during the US credit rating downgrade. The August 2015 equity flash crash pushed the VIX to 40.7. The February 2018 Volmageddon event, in which leveraged short-volatility exchange-traded products collapsed, produced a VIX peak of 37.3. The COVID-19 pandemic drove the VIX to its sample-period high of 82.7 on March 16, 2020. The Federal Reserve's 2022 rate hiking cycle lifted the VIX to 34.8.

Over the full sample, the VRP averages 3.67% annually, with a median of 4.1% and a standard deviation of 5.33%. Implied volatility exceeds realized volatility on 85.3% of all trading days, consistent with the empirical literature placing this frequency in the 75 to 85 percent range. Full analysis code and data pipeline are available at https://github.com/z-ghazanfar/spy-vrp-backtest.

Table 1: Summary Statistics

Metric	Value
Sample Period	Feb 2010 to Dec 2025
Trading Days	4,002
Mean IV (VIX)	18.4%
Mean RV (21-day)	14.73%
Mean VRP	3.67%
Median VRP	4.1%

Std. Dev. VRP	5.33%
% Days IV > RV	85.3%

4. Methodology

4.1 VRP Signal Construction

The core signal is the VRP, computed daily as the difference between the VIX and the trailing 21-day realized volatility of SPY log returns. Formally:

$$VRPt = IVt - RVt$$

where IVt is the VIX level on day t (annualized percentage points) and RVt is:

$$RVt = sqrt(252) * sigma(delta\ log\ St, ..., delta\ log\ St\text{-}21)$$

A positive VRP indicates that options are relatively expensive compared to recent historical volatility. The strategy enters a credit spread at month-end when VRPt exceeds 3%.

4.2 Credit Spread Strategy Design

On each monthly entry signal, the following bull put spread is simulated:

- Underlying: SPY (S&P 500 ETF)
- Short put: 2% out-of-the-money
- Long put: 4% out-of-the-money (defines maximum loss)
- Spread width: 2% of underlying price
- Net credit: 35% of spread width, used as a mid-range approximation of typical 20-30 DTE OTM put pricing. In practice this number varies with IV level: when VIX is low, 2% OTM puts pay well below 35%; when VIX is elevated, they pay more. The actual edge of the strategy scales with the VIX level, which is part of why the VRP filter helps. Transaction costs (commissions and bid-ask spread slippage) are not modeled and would reduce per-trade returns modestly; at roughly $1-2 per contract at major brokerages, they are minor relative to the spread premium collected.

The payoff function at expiration is as follows. If SPY declines less than 2%, the full premium is retained. If SPY declines between 2% and 4%, the loss is proportional to the intrinsic value of the short put. If SPY declines more than 4%, the maximum loss of 65% of spread width is incurred. Capital at risk per trade is assumed to be 20% of the portfolio, and returns are compounded as positions are closed and reopened monthly.

4.3 Signal Threshold Sensitivity Analysis

The choice of VRP threshold (3%) is evaluated systematically by computing the strategy win rate across thresholds from 0% to 12% in 0.5% increments. Higher thresholds improve win rates but reduce trade frequency, eventually producing insufficient sample sizes. The 3% threshold balances selectivity with trade frequency.

5. Results

5.1 The Variance Risk Premium: Persistence and Magnitude

The VRP exhibits strong persistence across the full sample period. Implied volatility exceeds realized volatility on 85.3% of trading days, with a mean VRP of 3.67% and a median of 4.1%. The positive skew in the distribution provides the statistical foundation for the strategy.

Figure 1 shows the time series of implied and realized volatility alongside the daily VRP. During calm markets, implied vol sits above realized vol by a moderate premium, which is the normal state. During stress events, realized volatility spikes sharply above implied vol for a brief window before normalizing. This VRP reversal is the most dangerous environment for short volatility strategies, and it's visible in all five annotated episodes.

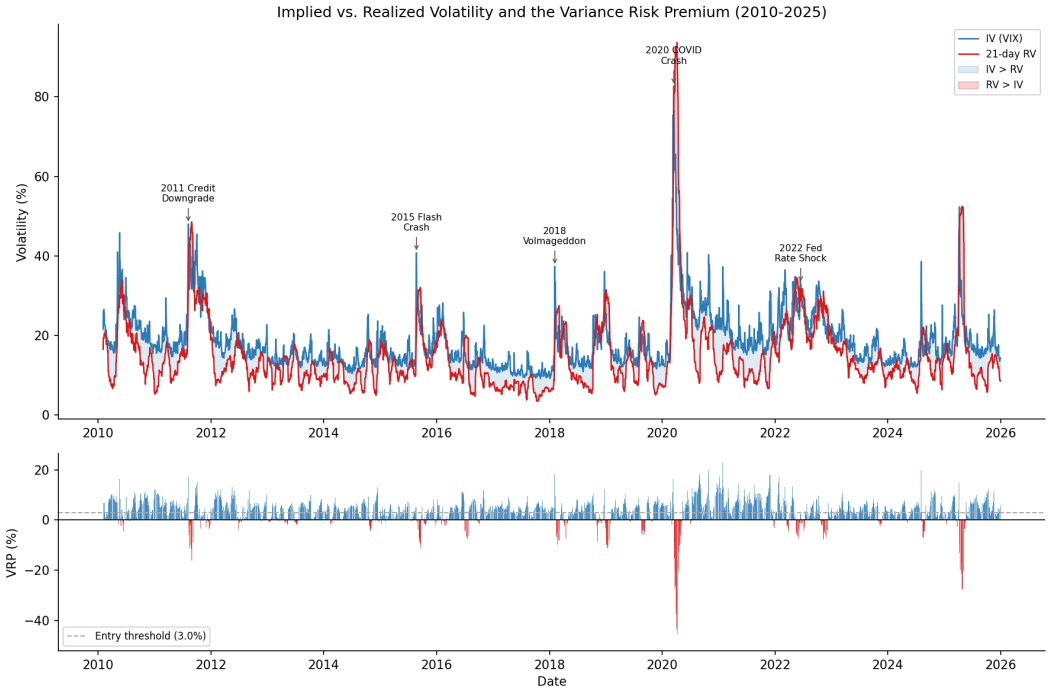

Figure 1: Implied volatility (VIX) and 21-day realized volatility over time (2010-2025). Blue shading indicates periods when IV > RV (the dominant regime); red shading indicates periods when RV > IV (stress episodes). Major volatility events are annotated. Bottom panel shows the daily VRP with the 3% entry threshold marked.

Figure 2 presents the VRP distribution and the predictive relationship between VRP quintiles and forward SPY returns. Higher VRP quintiles are generally associated with higher forward returns, consistent with the interpretation that elevated VRP signals risk aversion that resolves as a positive surprise. The lowest quintile (mean VRP of -3.36%, capturing post-crash environments when RV exceeds IV) shows the highest average forward return (1.42%), likely due to sharp recoveries following dislocations.

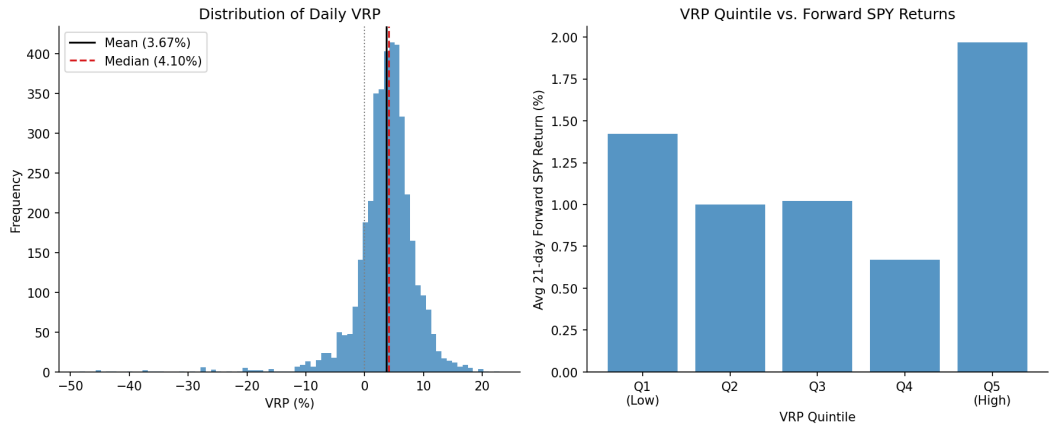

Figure 2: Left panel shows the distribution of daily VRP values with the mean and median marked. Right panel shows average 21-day forward SPY returns by VRP quintile, revealing a generally positive relationship between VRP level and subsequent returns.

5.2 Backtest Performance

The VRP-filtered strategy achieves a win rate of 86.0% across 121 monthly trades over the sample. On winning trades, the strategy captures an average of 34.67% of spread width, close to the full 35% premium, indicating that SPY rarely breaches the short strike. Losing trades average a 52.70% loss of spread width, suggesting losses are typically partial rather than maximal.

Under continuous reinvestment at 20% of portfolio per trade, the filtered strategy produces an annualized return of 37.5% with an annualized Sharpe ratio of 2.47. It's worth being clear about what's driving that number: the average trade generates 22.39% of spread width in P&L, which under a fixed-dollar allocation translates to roughly 4-5% annual portfolio return. The 37.5% figure reflects 16 years of compounding a growing percentage allocation, not the per-trade edge on its own. The always-on strategy achieves a higher annualized return of 56.2% by compounding more winning trades. The VRP filter reduces maximum portfolio drawdown significantly: -24.31% for the filtered strategy versus -40.71% for the always-on approach. The filter's primary value is therefore in risk reduction rather than return enhancement in this sample.

These results were generated during a historically favorable period for US equities. SPY returned approximately 14% annually from 2010 through 2025, and monthly declines exceeding 2% were relatively infrequent. The 86.0% win rate should be interpreted in this context: a period with more frequent large monthly drawdowns would reduce win rates and returns substantially.

Table 2: Backtest Performance Summary

Metric	Filtered (VRP > 3%)	Always-On
Monthly Trades	121	184
Win Rate	86.0%	85.3%
Avg P&L per Trade (% spread)	22.39%	N/A

Avg P&L: Wins	34.67%	N/A
Avg P&L: Losses	-52.70%	N/A
Annualized Return	37.5%	56.2%
Max Drawdown	-24.31%	-40.71%
Annualized Sharpe	2.47	N/A

Figure 3: Backtest results for the VRP-filtered strategy. Top-left: cumulative portfolio P&L comparison between the filtered and always-on strategies under continuous 20% reinvestment. Top-right: monthly trade P&L by outcome. Bottom-left: scatter of VRP at entry versus SPY return over the trade period, with short and long strike levels marked. Bottom-right: drawdown profile of the filtered strategy over time.

5.3 Tail Risk and Stress Analysis

Figure 4 examines tail risk from two angles. The left panel compares the average VRP before versus during five major stress events. In four of the five cases (2011, 2015, 2018, 2020), the VRP was positive and elevated before the event, then turned sharply negative during it as realized volatility spiked far above implied vol. The COVID-19 episode is the most extreme example, with the average VRP during the event window falling to approximately -12%, consistent with the VIX reaching 83 while SPY was already in freefall. The 2022 rate shock is the exception: the VRP stayed positive throughout, because the Federal Reserve's tightening cycle was slow and telegraphed rather than a

sudden crash, so options markets priced in elevated vol before realized vol fully caught up.

This dynamic has a direct implication for strategy design. Positions entered during high-VRP environments are more likely to have been entered when premium was adequate. Entering positions when the VRP is low or negative, a signal of a market already in distress, is when the strategy is most exposed. The VRP threshold filter directly addresses this by excluding entries during low-premium regimes.

The right panel shows win rate sensitivity to the entry threshold. At a threshold of 0%, the win rate is already around 85.3%, reflecting the underlying high base rate of profitable months across the sample. Raising the threshold to 3% produces a marginally higher 86.0% win rate while meaningfully reducing trade count from 184 to 121. At thresholds above 7%, win rates become highly variable as sample sizes shrink below 30 trades. The 3% threshold was selected as the point that captures the base-rate win rate advantage without sacrificing so many trades that the compounding benefit disappears. The maximum drawdown of -24.31% occurred during the COVID-19 episode when two consecutive months saw SPY fall beyond the short strike, generating back-to-back near-maximum losses.

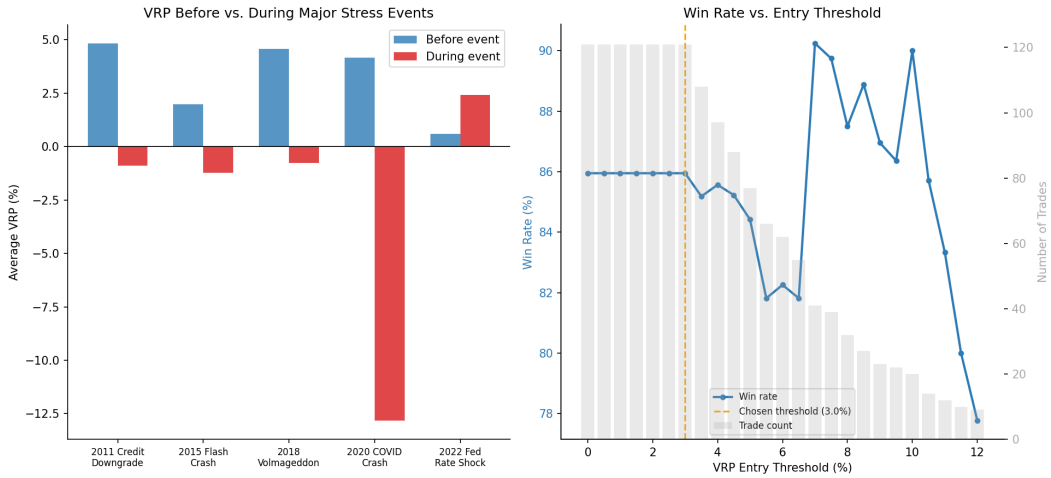

Figure 4: Left panel shows average VRP before versus during five major stress events, illustrating the VRP reversal dynamic in which realized volatility spikes sharply above implied volatility during dislocations. Right panel shows win rate as a function of VRP entry threshold, with the chosen 3% threshold marked and trade count overlaid.

6. Practical Deployment Framework

The strategy described above can be implemented with a standard retail brokerage account and limited capital. This section outlines the practical setup.

6.1 Brokerage and Account Requirements

The strategy requires an options-enabled brokerage account with at least Level 2 options approval. Brokerages such as Tastytrade, TD Ameritrade (thinkorswim), and Interactive Brokers are well-suited due to their low per-contract commissions and intuitive spread

order entry. A minimum account of $3,000 to $5,000 is recommended to allow position diversification and to absorb a losing month without distress.

6.2 Signal Monitoring and Trade Entry

The VRP signal can be monitored using freely available data. The VIX is published continuously by the CBOE during market hours; trailing 21-day realized volatility can be computed from daily SPY closing prices on Yahoo Finance. The entry signal triggers when VRP = VIX - RV21 > 3%.

On the last trading day of each month when the signal is active, enter a bull put spread on SPY: short put approximately 2% below current price, long put approximately 4% below current price, targeting 20 to 30 days to expiration and collecting at least 30% of spread width as premium. No single spread should exceed 20% of account value.

6.3 Risk Management

Maximum loss on any single trade is defined at entry. If SPY declines to within 1% of the short strike mid-trade, closing early for a partial loss is preferable to holding through expiration. Pausing the strategy when the VIX exceeds 35 is prudent, as the VRP signal becomes less reliable during extreme volatility regimes. The -24.31% maximum drawdown from the backtest illustrates that two consecutive losing months at near-maximum loss is a realistic tail scenario that investors should plan for before entering the strategy.

7. Conclusions

This paper documents the existence and persistence of the variance risk premium in equity index options using approximately 16 years of real SPY and VIX data. Over the full sample (2010 to 2025), implied volatility exceeded realized volatility on 85.3% of trading days with a mean VRP of 3.67%, providing a consistent structural edge for options sellers.

By conditioning entry on a minimum VRP threshold of 3%, the strategy achieves an 86.0% win rate across 121 monthly trades and an annualized Sharpe ratio of 2.47. The primary benefit of the filter relative to always-on trading is risk reduction: the maximum portfolio drawdown falls from -40.71% to -24.31%. However, the always-on strategy achieves a higher annualized return (56.2% versus 37.5%) by compounding more trades, which reflects a fundamental tradeoff between signal selectivity and return.

The tail risk analysis reveals the strategy's key structural vulnerability: during severe dislocations, realized volatility spikes sharply above implied volatility, generating near-maximum losses on active positions. This suggests supplementing the VRP threshold with additional macro risk monitoring, for example suspending positions when the VIX itself exceeds a stress threshold or when the VRP turns negative.

The sample period (2010 to 2025) was historically favorable for US equities, and the win rates reflect this. A longer or more turbulent sample would likely produce lower win rates and returns. Future extensions could incorporate term structure signals (VIX3M/VIX ratio), put/call ratios, or regime-conditional analysis across high- and low-volatility environments to improve robustness out of sample.

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